

TECHNICAL REPORT: VIBE-GUARD AI

1. Project Abstract

Vibe-Guard AI is a decentralized, privacy-preserving intelligence system designed to mitigate 'Decision Fatigue' and emotional burnout in real-time. By utilizing Federated Learning (FL) for decentralized intelligence and FedWAPR for robust model aggregation, the system provides users with cognitive 'guardrails.' It prevents high-stakes errors in communication and financial decision-making by monitoring the user's subconscious emotional state locally on their device.

2. Problem Statement

In 2026, cognitive overload is a primary bottleneck to human productivity. Continuous digital interaction leads to subconscious burnout, causing individuals to make poor decisions. Current solutions are either intrusive (cloud-based data logging) or reactive. There is a critical need for a proactive, decentralized AI that protects cognitive health while ensuring absolute data privacy.

3. Methodology: FedWAPR and Pathological Saliency

1. FedWAPR (2025): Implements Weighted Aggregation via Probability-based Ranking. It ranks model updates to ensure only reliable behavioral patterns influence the global emotional baseline, neutralizing noisy or fraudulent sensor data.
2. Pathological Saliency (2023): Analyzes vocal and behavioral biomarkers (jitter, shimmer, typing cadence) locally to detect micro-stresses invisible to the human ear.
3. Explainable AI (2025): Uses attention mechanisms to provide heatmaps and explanations to the user when a cognitive guardrail is triggered, ensuring transparency.

4. Technical Architecture

- Edge Intelligence: React Native + TensorFlow Lite (LiteRT) for on-device backpropagation.
- Coordination: Flower (flwr.dev) manages weight exchange between the edge and the server.
- Backend: Python (FastAPI) for FedWAPR-based weight aggregation and global model synchronization.

5. Impact and Revenue

Vibe-Guard AI serves a dual market: a B2C subscription for individual 'Cognitive Insurance' and a B2B enterprise model for high-pressure corporate environments. It generates high value by reducing employee burnout and providing institutional-level anonymized sentiment heatmaps.